Anndery Lopes

Senior Software Engineer

Barbacena, MG, Brazil, 36205501 | sensacional157@hotmail.com | https://www.linkedin.com/in/anndery-lopes-732864214/

Summary

As a Senior Full-Stack and AI Engineer with 7+ years of experience, I specialize in designing and delivering high-performance, distributed systems using .NET, Node.js, and modern web technologies, with a strong focus on AI-driven applications and large-scale data acquisition pipelines. I have led end-to-end development of real-time, high-throughput platforms integrating machine learning models, event-driven architectures, and scalable APIs across cloud-native environments. My experience spans building intelligent automation systems, data ingestion frameworks, and full-stack applications leveraging React, TypeScript, and microservices architectures on AWS. At Boston Consulting Group and Jelvix, I worked on enterprise-grade AI solutions, internal tooling, and data platforms that support decision-making at scale, while my academic and early industry work focused on backend systems, distributed processing, and data collection frameworks. I bring deep expertise in system design, low-latency architectures, and production-grade AI integration, with strong ownership across the full software lifecycle from architecture through deployment and optimization.

Technical Skills

- Programming Languages: C#, TypeScript, JavaScript (ES6+), Python, SQL, Bash, PowerShell
- Frameworks & Runtime: .NET Core, ASP.NET Core, Entity Framework Core, Node.js, Express.js, NestJS, React, Next.js, Redux Toolkit
- AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenAI APIs, NLP, Deep Learning, Model Training, Model Serving, Feature Engineering, Data Preprocessing, MLOps fundamentals
- Data Engineering & Acquisition: Data Acquisition Pipelines, Web Scraping, ETL/ELT, Apache Kafka, RabbitMQ, Data Streaming, Batch Processing, Data Validation, Data Normalization, Airflow (concepts), Pandas
- Frontend Technologies: React, Next.js, TypeScript, HTML5, CSS3, Tailwind CSS, Material UI, Responsive Design, Accessibility (WCAG), Performance Optimization
- Backend & APIs: RESTful APIs, GraphQL, gRPC (concepts), Microservices Architecture, Event-Driven Architecture, Domain-Driven Design (DDD), API Gateway Patterns, Rate Limiting
- Databases & Storage: PostgreSQL, SQL Server, MongoDB, Redis, Elasticsearch, Data Modeling, Query Optimization, Indexing Strategies, Read/Write Splitting
- Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS, CloudWatch, IAM), Docker, Kubernetes, Nginx, Serverless Architecture, Infrastructure as Code (Terraform – concepts)
- DevOps & CI/CD: GitHub Actions, Azure DevOps, CI/CD Pipelines, Docker Compose, Build Automation, Release Management
- Messaging & Streaming: Apache Kafka, RabbitMQ, Event Streaming, Pub/Sub Systems
- Testing & Quality: xUnit, NUnit, Jest, Mocha, Integration Testing, Unit Testing, Test-Driven Development (TDD)
- Observability & Performance: Logging (Serilog, Winston), Monitoring, Distributed Tracing, Profiling, Performance Tuning
- Security: OAuth2, JWT, Role-Based Access Control (RBAC), API Security Best Practices
- Tools & Platforms: Git, GitHub, GitLab, Postman, Swagger/OpenAPI, Linux, VS Code, Visual Studio

Professional Experience

Senior Software Engineer

Boston Consulting Group, Boston, Massachusetts 04/2024 – Present

- Architected and led the development of a large-scale AI-driven decision intelligence platform using .NET Core, Node.js, and Python, designing a microservices-based system that processes high-throughput enterprise datasets and exposes low-latency APIs consumed by internal analytics tools and client-facing dashboards, ensuring scalability through containerization with Docker and orchestration using Kubernetes
- Designed and implemented complex data acquisition pipelines using Python, Apache Kafka, and AWS services (S3, Lambda, RDS), enabling ingestion of structured and unstructured data from third-party APIs, web scraping systems, and internal data sources, while building robust data validation, transformation, and enrichment layers to support downstream machine learning workflows
- Engineered event-driven architectures using Kafka and asynchronous messaging patterns, enabling real-time stream processing and decoupled communication between distributed services, significantly improving system resilience and throughput under concurrent workloads
- Developed high-performance RESTful and GraphQL APIs using ASP.NET Core and Node.js, incorporating Redis caching, query optimization, and connection pooling strategies to handle high request volumes with minimal latency
- Integrated machine learning models into production-grade systems using TensorFlow and PyTorch, building scalable inference services and model-serving APIs, while implementing versioning, monitoring, and fallback mechanisms to ensure reliability in production environments
- Designed scalable microservices architecture with clear domain boundaries using Domain-Driven Design principles, enabling independent deployment and horizontal scaling of services across cloud environments

- Optimized database performance across PostgreSQL and SQL Server by implementing advanced indexing strategies, query refactoring, and read/write separation patterns, reducing query execution time for large datasets
- Built sophisticated frontend applications using React, Next.js, and TypeScript, implementing advanced state management, real-time data updates via WebSockets, and performance optimizations such as lazy loading and memoization
- Developed CI/CD pipelines using GitHub Actions, Docker, and Kubernetes, automating build, test, and deployment processes while ensuring zero-downtime deployments and consistent environment configurations
- Implemented secure authentication and authorization mechanisms using OAuth2, JWT, and RBAC, ensuring compliance with enterprise-grade security standards
- Designed and deployed API gateway solutions using Nginx and custom middleware to centralize routing, authentication, and rate limiting across distributed services
- Enhanced observability by integrating structured logging (Serilog), distributed tracing, and monitoring tools, enabling proactive identification of performance bottlenecks and system failures
- Reduced latency in critical data processing paths by implementing parallel processing, asynchronous execution, and efficient data structures within high-throughput pipelines
- Built reusable internal SDKs and shared service libraries to standardize communication patterns and accelerate development across multiple teams
- Collaborated with data scientists and product stakeholders to translate complex business requirements into scalable AI-powered solutions
- Implemented fault tolerance mechanisms including retries, circuit breakers, and graceful degradation strategies to ensure high system availability
- Developed real-time analytics dashboards using React and WebSockets to visualize streaming data and key system metrics
- Mentored engineers on distributed systems, system design, and best practices, contributing to team technical excellence
- Led migration initiatives from monolithic architectures to microservices-based systems, improving scalability and maintainability
- Conducted deep performance profiling and optimization across backend services, significantly improving throughput and system efficiency

Software Engineer

Galena, Sao Paulo, Sao Paulo 08/2021 – 03/2024

- Architected and delivered enterprise-grade full-stack platforms using .NET Core, ASP.NET Core, React, and TypeScript, designing microservices-based systems with clearly defined service boundaries, implementing business logic layers and API contracts, and deploying containerized services using Docker to ensure scalability, maintainability, and consistent environments across development and production
- Designed and implemented backend services using ASP.NET Core and Entity Framework Core, building high-throughput RESTful APIs with advanced validation, middleware pipelines, and exception handling mechanisms, while optimizing performance through asynchronous programming patterns and efficient database interaction strategies
- Developed AI-powered modules using Python, Scikit-learn, and Pandas, training and deploying predictive models, exposing inference endpoints via REST APIs, and integrating model outputs into .NET and Node.js services to enable intelligent automation and decision-making workflows within production applications
- Engineered robust data acquisition frameworks using Node.js, Python, and message brokers such as RabbitMQ, building pipelines that ingest high-frequency data from third-party APIs and IoT devices, implementing buffering, batching, retry logic, and schema validation to ensure data consistency and fault tolerance
- Optimized backend performance by implementing Redis-based caching strategies, connection pooling, and query optimization techniques in PostgreSQL, significantly improving response times for data-intensive endpoints under concurrent user load
- Built scalable frontend applications using React, Next.js, and TypeScript, designing reusable component architectures, implementing state management with Redux Toolkit, and optimizing rendering performance through memoization, lazy loading, and efficient component lifecycle management
- Leveraged AWS cloud services including EC2, S3, and RDS to design and deploy scalable infrastructure, configuring auto-scaling groups, storage layers, and managed databases to support high-availability production systems
- Implemented event-driven architectures using RabbitMQ and asynchronous messaging patterns, enabling decoupled communication between microservices and improving system resilience and scalability under peak workloads
- Improved database performance in PostgreSQL by redesigning schemas, implementing indexing strategies, and optimizing complex queries, reducing execution time for large datasets and improving overall system throughput
- Automated build and deployment workflows using Docker, GitHub Actions, and CI/CD pipelines, ensuring consistent builds, automated testing, and reliable deployment processes across multiple environments
- Designed and implemented secure authentication and authorization mechanisms using JWT and OAuth2, incorporating role-based access control and secure token handling to protect application endpoints
- Built comprehensive logging and monitoring systems using structured logging frameworks and centralized logging pipelines, enabling real-time visibility into system behavior and faster issue resolution
- Reduced latency in data processing pipelines by implementing parallel execution models and efficient data transformation techniques using Python and Node.js

- Collaborated closely with product managers and designers to translate business requirements into scalable technical solutions, ensuring alignment between system architecture and user needs
- Developed reusable backend modules and shared services using .NET Core and Node.js, improving development efficiency and consistency across multiple projects
- Implemented real-time communication features using WebSockets and Node.js, enabling live updates and notifications for end users
- Conducted load testing and performance benchmarking using realistic traffic scenarios, identifying bottlenecks and optimizing system components to meet scalability requirements
- Mentored junior engineers on full-stack development, distributed systems, and best practices, contributing to team knowledge sharing and technical growth
- Contributed to architectural decisions including service decomposition, data flow design, and infrastructure planning, ensuring long-term scalability and maintainability
- Participated in code reviews and enforced coding standards, improving overall code quality and maintainability across the codebase

Software Engineer

Universidade Federal de Minas Gerais, Belo Horizonte, Minas Gerais 2/2019 – 06/2021

- Developed backend systems using .NET Core and ASP.NET Core, designing modular and maintainable services that expose RESTful APIs, implementing middleware pipelines, validation layers, and structured error handling to support reliable web-based platforms
- Built data acquisition tools using C#, Python, and Node.js to capture user interactions and system metrics, designing pipelines that perform data collection, transformation, and storage, ensuring data consistency and readiness for analytics
- Designed and implemented RESTful APIs with ASP.NET Core, incorporating routing, model binding, and validation, while optimizing request handling through asynchronous programming and efficient resource utilization
- Modeled and optimized relational databases in SQL Server, designing normalized schemas, implementing indexing strategies, and tuning queries to improve performance for transactional and analytical workloads
- Developed frontend components using React and modern JavaScript, implementing dynamic UI elements, responsive layouts, and efficient rendering techniques to enhance user experience
- Integrated third-party APIs using HTTP clients and SDKs, handling authentication, rate limiting, and error scenarios to extend platform functionality
- Implemented asynchronous job processing systems using background workers and task queues, enabling efficient handling of long-running operations and improving overall system responsiveness
- Identified and resolved performance bottlenecks through profiling tools and code analysis, optimizing CPU and memory usage across backend services
- Built authentication and authorization systems using JWT and secure token-based mechanisms, ensuring proper access control and protection of sensitive endpoints
- Implemented logging frameworks using structured logging approaches, capturing detailed application metrics and supporting debugging and monitoring workflows
- Participated in system design discussions, contributing to architectural improvements and scalability considerations for growing applications
- Developed data processing workflows using Python and C#, transforming raw data into structured formats suitable for storage and analysis
- Implemented caching strategies using Redis to reduce database load and improve API response times for frequently accessed data
- Wrote comprehensive unit and integration tests using xUnit and other testing frameworks, ensuring code reliability and maintainability
- Collaborated with cross-functional teams including designers and product managers to deliver features aligned with business goals
- Maintained and refactored legacy systems, improving code structure, readability, and scalability
- Diagnosed and resolved production issues using logging, monitoring, and debugging tools, ensuring minimal downtime
- Developed reusable components and utilities to standardize development patterns and reduce duplication
- Supported deployment processes using CI/CD pipelines and environment configuration tools
- Documented system architecture, APIs, and technical decisions to support knowledge sharing and onboarding

Education

Universidade Federal de Minas Gerais

Bachelor's degree, Computer Science | Belo Horizonte, Minas Gerais |08/2017 – 07/2021